Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058, India

(Autonomous College Affiliated to University of Mumbai)

End Semester Examination
Set - II

June 2025

Max. Marks: 100 | **Duration: 3 Hrs**

Class: TE CSE(DS/AIML) | **Semester: VI**

Course Code: DS306/AI306 | **Program: B.Tech**

Name of the Course: Distributed Computing

Instruction:

(1) All questions are compulsory

(2) Draw neat diagrams and write equations wherever necessary

(3) Assume suitable data if necessary

Course Outcomes:At the end of the course student will be able to:

(1) Understand the principles and desired properties of distributed systems. Apply the various communication techniques for distributed communication.

(2) Apply the concepts of process, naming, consistency, replication and faults tolerance in a distributed environment.

(3) Apply the algorithms such as clock synchronization, election, and mutual exclusion in distributed applications.

(4) Identify the challenges in developing distributed applications.

Q No.		Max. Marks	CO-BL
Q.1a	Solve the following clock synchronization problems (5 marks each): 1. Cristian's Algorithm: A client machine sends a time request to a time server at *T1 = 10:00:00.000 (local time)*. The server responds at *T2 = 10:00:00.150 (server time)*, and the client receives the response at *T3 = 10:00:00.300 (local time)*. Assuming negligible processing time at the server, calculate: 1. The round-trip time (RTT). 2. The adjusted client time after synchronization. 2. Berkeley's Algorithm: In a distributed system, a master polls 3 slaves for their clock differences: Slave A: Reports +50 ms (ahead of master). Slave B: Reports -30 ms (behind master). Slave C: Reports +10 ms (ahead of master). Calculate: 1. The average clock deviation. 2. The final synchronized time adjustment applied to all nodes.	10	CO3-BL3
Q.1b	Why is scalability an important feature in the design of a distributed system? Discuss some of the guiding principles for designing a scalable distributed system.	10	CO4-BL2
Q.2a	Scenario: An airline reservation system uses Message-Oriented Middleware (MOM) (e.g., RabbitMQ, Kafka) to process seat bookings from multiple travel agencies. Explain how MOM ensures reliable and asynchronous booking transactions.	10	CO1-BL3

Q2.b	Compare and contrast shared-memory and message-passing architectures in parallel computing. Discuss their advantages, limitations, and real-world applications with suitable examples. OR Discuss the challenges of implementing a Distributed Operating System (DOS) in real-world environments. How do these challenges differ from those in Network Operating Systems (NOS)?	10	CO1-BL2
Q.3a	What is a datagram, and why are multi-datagram messages important for communication between processes (IPC)? What are the challenges associated with handling multi-datagram messages in such systems, and what mechanisms can be used to address these challenges?	10	CO4-BL3
Q.3b	In a client-server IPC, a client sends a request message to a server, and the server processes the request and returns the result of the request processing to the client. Is it useful for a process to behave both as a client and a server? If no, explain why. If yes, give an example in support of your answer. OR Describe some flexibility features that a message-passing system should provide to its users. Write suitable IPC primitives that will allow the users to take advantage of these flexibility features.	10	CO2-BL2
Q.4a	Scenario: An e-commerce platform (e.g., Amazon) uses multiple servers to handle checkout requests during a flash sale. Session persistence is critical (users must stay connected to the same server for cart consistency). 1. Should static or dynamic load balancing be used? Why? 2. Which strategy (Round Robin, Weighted Round Robin, or IP Hashing) is best? Justify. 3. What is a limitation of your chosen approach?	10	CO2-BL3
Q.4b	In a distributed system, three processes (P1, P2, P3) exchange messages. Their local events and message exchanges are logged as follows: P1: Event A (local), Sends M1 to P2 P2: Receives M1, Event B (local), Sends M2 to P3 P3: Receives M2, Event C (local) Task: Assign Lamport timestamps to each event (A, B, C, M1, M2). Assume initial clocks = 0. Determine if the events are causally ordered.	10	CO3-BL4
Q.5a	In a distributed system like Amazon DynamoDB or Dropbox, data is replicated across multiple servers to ensure availability and fault tolerance. However, due to network delays or failures, clients may observe inconsistent data. Using a real-time example such as a user updating their profile picture in a cloud-based social media application (e.g., Facebook or Instagram), explain the following client-centric consistency models: 1. Read-Your-Writes Consistency 2. Monotonic Reads Consistency 3. Monotonic Writes Consistency 4. Eventual Consistency How does each model affect the user experience in this scenario?	10	CO3-BL3

Q.5b	Could you elaborate on the concept of causal consistency within distributed systems? Furthermore, in the context of real-time collaborative applications, such as a multi-user document editing platform, where users concurrently modify shared content, why would causal consistency be the most appropriate consistency model to ensure both logical ordering and an intuitive user experience?	10	CO2-BL3

Course outcome	**Marks in ESE**	Marks in MSE	Total marks for each course outcome
CO1	20	10	30
CO2	30	10	40
CO3	30	5	35
CO4	20	5	25